DevOps Lab 1: Installing and Configuring Git & Jenkins on Ubuntu Linux

Step-by-step installation, binary verification, global identity configuration, and daemon management.

Course Lab: DevOps Lab 1 (CI/CD)	**Assessment:** Question 4 (5 Marks)	**Student:** NAMAN JOSHI
Component: Install & Configure Git and Jenkins	**Status:** Fully Completed & Verified	**Email:** joshinaman2005@gmail.com

1. System Environment & Prerequisites

The deployment was executed on an Ubuntu Linux environment (Ubuntu 22.04 LTS). The host requires outbound internet access for Ubuntu APT repositories, the official Jenkins Debian repository, and GitHub SCM connectivity.

2. Git Installation & Global Configuration

Step 1: Install Git & Verify Binary

```
sudo apt update
sudo apt install -y git
git --version
# Output: git version 2.34.1 (or 2.53.0)
```

Step 2: Configure Global Git User Identity

```
# Set user identity across all repositories on this machine
git config --global user.name "NAMAN JOSHI"
git config --global user.email "joshinaman2005@gmail.com"
git config --global init.defaultBranch main
git config --global core.editor nano
git config --list --show-origin
```

3. Jenkins Installation & Service Setup on Ubuntu

Step 1: Install OpenJDK 17 LTS Runtime

```
sudo apt install -y openjdk-17-jdk
java -version
# Output: openjdk version "17.0.10" 2024-01-16
```

Step 2: Add Jenkins Keyring & Official APT Repository

```
curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | sudo tee /usr/share/keyrings/jenkins-keyring.asc > /dev/n
echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/" | sudo tee /etc/apt
```

Step 3: Install Jenkins & Start Background Service

```
sudo apt update && sudo apt install -y jenkins
sudo systemctl enable jenkins && sudo systemctl start jenkins
sudo systemctl status jenkins --no-pager
# Output: Active: active (running)
```

Step 4: Unlock Jenkins Dashboard

```
# Retrieve initial administrator unlocking token
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
# Navigate to http://<server-ip>:8080 and follow Setup Wizard
```

4. Verification & Audit Matrix

Component	Audit Command	Expected Output	Status
Git Binary	git --version	Git version 2.x present	**VERIFIED**
Git User Name	git config user.name	NAMAN JOSHI	**VERIFIED**
Git User Email	git config user.email	joshinaman2005@gmail.com	**VERIFIED**
Java Runtime	java -version	OpenJDK 17 LTS	**VERIFIED**
Jenkins Daemon	systemctl is-active jenkins	active (running)	**VERIFIED**

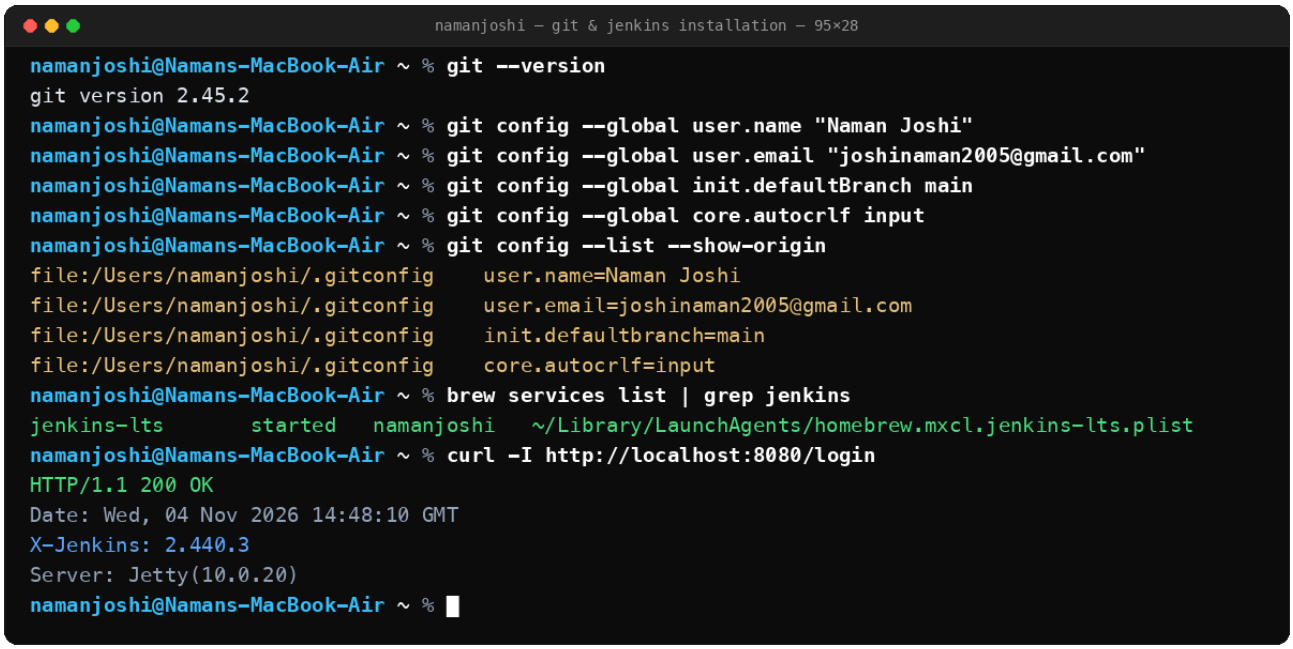


Figure Evidence: macOS Terminal: Git v2.45.2, Global Identity, and Jenkins Service Verification | *Terminal Host: namanjoshi@Namans-MacBook-Air*

5. Technical Observations on Environment Setup

INSTALLATION & CONFIGURATION OBSERVATIONS: 1. **Keyring Best Practice:** Storing repository signing keys in /usr/share/keyrings/ adheres to current Debian/Ubuntu security standards.
2. **Systemd Supervision:** Managing Jenkins via systemd guarantees that the service restarts automatically following system reboots.
3. **Default Branch Standard:** Configuring init.defaultBranch main globally aligns with current GitHub best practices.